

BASE ORCADAS, Antarctica

WMO: 889680

Lat: 60.7378S

Lon: 44.7383W

Elev: 8

StdP: 101.23

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-26.4	-24.1	-30.6	0.2	-26.2	-27.4	0.3	-23.9	21.5	-5.0	20.7	-5.1	5.4	140	1.077

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	5.7	3.8	4.7	3.0	4.0	2.5	4.1	5.3	3.4	4.3	2.8	3.6	7.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.1	4.7	4.2	2.5	4.5	3.5	2.1	4.4	2.9	17.2	5.4	15.8	4.5	14.7	3.7	9.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	15.1	13.2	DB	-29.2	8.5	2.1	1.5	-30.7	9.6	-31.9	10.4	-33.1	11.3	-34.6	12.4	(4)
(5)			WB	-29.2	6.2	1.9	1.1	-30.6	7.0	-31.7	7.7	-32.8	8.3	-34.2	9.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	1.3	1.3	0.5	-1.8	-4.7	-7.1	-9.6	-8.6	-6.0	-2.4	-0.5	0.7	(6)
(7)		DBStd	5.94	1.36	1.49	2.09	3.22	5.20	6.10	7.48	6.98	5.98	3.90	2.32	1.43	(7)
(8)		HDD10.0	4781	269	244	294	354	457	512	608	576	481	384	314	289	(8)
(9)		HDD18.3	7822	527	477	552	604	715	762	866	834	731	643	564	547	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.5	5.8	6.4	6.9	6.7	6.6	6.3	6.5	6.7	6.6	6.8	6.4	6.0	(14)
(15)	Precipitation	PrecAvg	803	70	77	105	84	60	47	50	55	58	74	67	55	(15)
(16)		PrecMax	997	117	123	178	124	130	79	96	104	99	145	116	97	(16)
(17)		PrecMin	583	45	49	64	46	25	23	16	9	21	20	25	16	(17)
(18)		PrecStd	108	19	16	26	22	23	17	22	27	24	25	24	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.7	6.9	5.2	4.3	2.8	2.2	2.8	4.7	4.7	5.8	6.2	(19)
(20)			MCWB	4.6	4.7	4.9	3.4	2.5	1.5	0.7	1.3	2.8	2.8	3.6	3.8	(20)
(21)		2%	DB	4.9	5.1	5.1	3.2	2.5	1.2	0.8	1.2	2.6	3.3	4.3	4.4	(21)
(22)			MCWB	3.2	3.5	3.5	1.9	1.4	0.4	0.2	0.5	1.3	1.8	2.4	2.5	(22)
(23)		5%	DB	3.9	4.0	4.0	2.2	1.5	0.4	0.1	0.4	1.4	2.3	3.1	3.2	(23)
(24)			MCWB	2.6	2.7	2.8	1.4	0.8	-0.2	-0.3	-0.1	0.5	1.0	1.7	1.8	(24)
(25)		10%	DB	3.2	3.2	3.1	1.6	0.9	-0.3	-0.4	-0.3	0.6	1.4	2.2	2.6	(25)
(26)			MCWB	2.1	2.2	2.2	1.0	0.4	-0.8	-0.8	-0.7	-0.1	0.5	1.1	1.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	5.1	5.0	3.7	2.8	1.6	1.2	1.5	3.1	3.1	4.0	4.1	(27)
(28)			MADB	6.3	6.3	6.7	4.7	3.8	2.5	1.6	2.1	4.2	4.2	5.3	5.7	(28)
(29)		2%	WB	3.6	3.7	3.8	2.2	1.6	0.6	0.2	0.6	1.5	2.0	2.7	2.8	(29)
(30)			MADB	4.5	4.6	4.7	2.8	2.2	1.0	0.5	1.1	2.3	3.0	3.9	4.0	(30)
(31)		5%	WB	2.8	3.1	3.0	1.7	1.0	0.0	-0.2	0.0	0.6	1.2	1.9	2.1	(31)
(32)			MADB	3.5	3.7	3.7	2.1	1.4	0.3	0.0	0.3	1.1	1.9	2.8	3.0	(32)
(33)		10%	WB	2.3	2.5	2.3	1.1	0.4	-0.7	-0.8	-0.6	0.0	0.7	1.3	1.6	(33)
(34)			MADB	3.0	3.1	2.9	1.6	0.8	-0.3	-0.4	-0.3	0.5	1.2	2.0	2.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.8	2.6	3.0	3.4	5.0	6.1	7.4	7.1	6.5	4.7	3.9	3.1	(35)
(36)			MADB	4.1	3.9	4.2	3.7	4.3	5.1	6.0	5.8	5.4	4.7	4.9	4.3	(36)
(37)			MCWBR	3.0	3.1	3.3	3.2	3.7	4.8	5.9	5.6	4.7	3.8	3.8	3.1	(37)
(38)		5% WB	MADB	3.7	3.6	4.0	3.6	3.9	4.9	5.9	5.6	5.2	4.3	4.6	3.9	(38)
(39)			MCWBR	3.0	3.1	3.3	3.2	3.6	4.8	5.9	5.5	4.7	3.7	3.7	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.314	0.302	0.284	0.229	0.179	0.209	0.251	0.277	0.302	0.295	0.305	(40)	
(41)		taud	2.497	2.524	2.527	2.437	2.193	1.991	2.118	2.270	2.326	2.349	2.429	2.465	(41)	
(42)		Ebn at Noon	964	920	850	707	520	406	525	721	853	918	980	988	(42)	
(43)		Edh at Noon	99	87	72	53	34	24	34	59	84	102	105	105	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.50	3.18	1.82	0.92	0.34	0.16	0.26	0.91	2.18	3.74	5.21	5.43	(44)	
(45)		RadStd	0.33	0.28	0.09	0.09	0.04	0.01	0.02	0.09	0.20	0.44	0.65	0.55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page